



Application Note

# **Power Measurement On the Ensemble E8 DevKit (DK-E8)**

**Version 1.0**

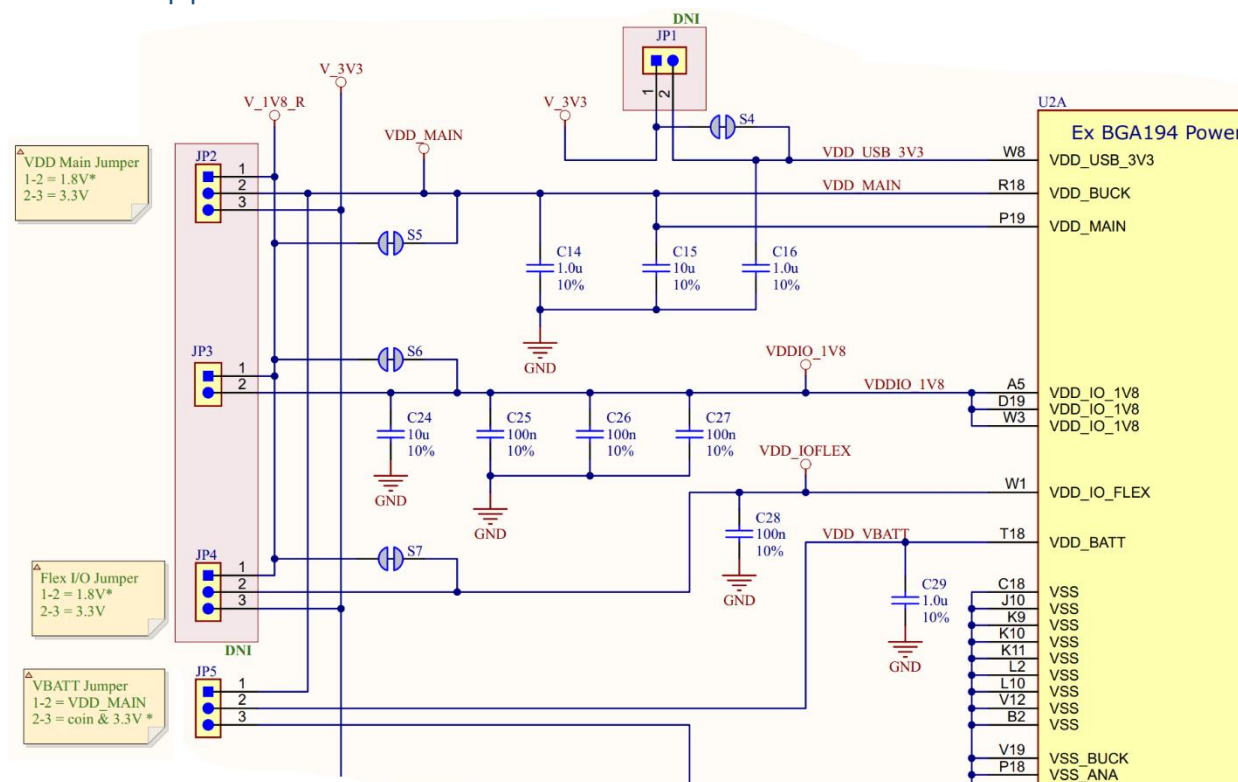
## Table of Contents

|                                |   |
|--------------------------------|---|
| Board Version.....             | 3 |
| Device Supplies .....          | 3 |
| Power Measurement Points ..... | 3 |
| Board Modifications.....       | 4 |
| Document History .....         | 5 |

## Board Version

This application note is applicable to the Ensemble E8 DevKit (DK-E8) Rev. A or Rev. B

## Device Supplies



## Power Measurement Points

The DevKit has no shunt resistors so an ammeter or DMM with ammeter mode is required.

JP1 allows you to measure VDD\_USB\_3V3. This supply is only relevant if the USB PHY is to be used, otherwise it can be left floating or tied to ground to reduce power.

JP2 allows you to measure VDD\_BUCK, VDD\_MAIN, and VDD\_BATT.

JP3 allows you to measure VDD\_IO\_1V8.

JP4 allows you to measure and configure VDD\_IO\_FLEX for 1.8V or 3.3V mode.

JP5 should be kept in position 1-2 so that VDD\_BATT is tied to BUCK and MAIN on JP2.

Most power will be drawn through JP2 (BUCK+MAIN+BATT). This is the primary measuring point when evaluating the Ensemble MCU's power modes.

During heavy IO operations, there will be power drawn through JP3 (VDD\_IO\_1V8).

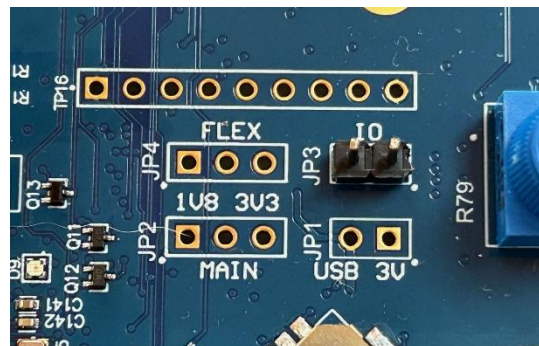
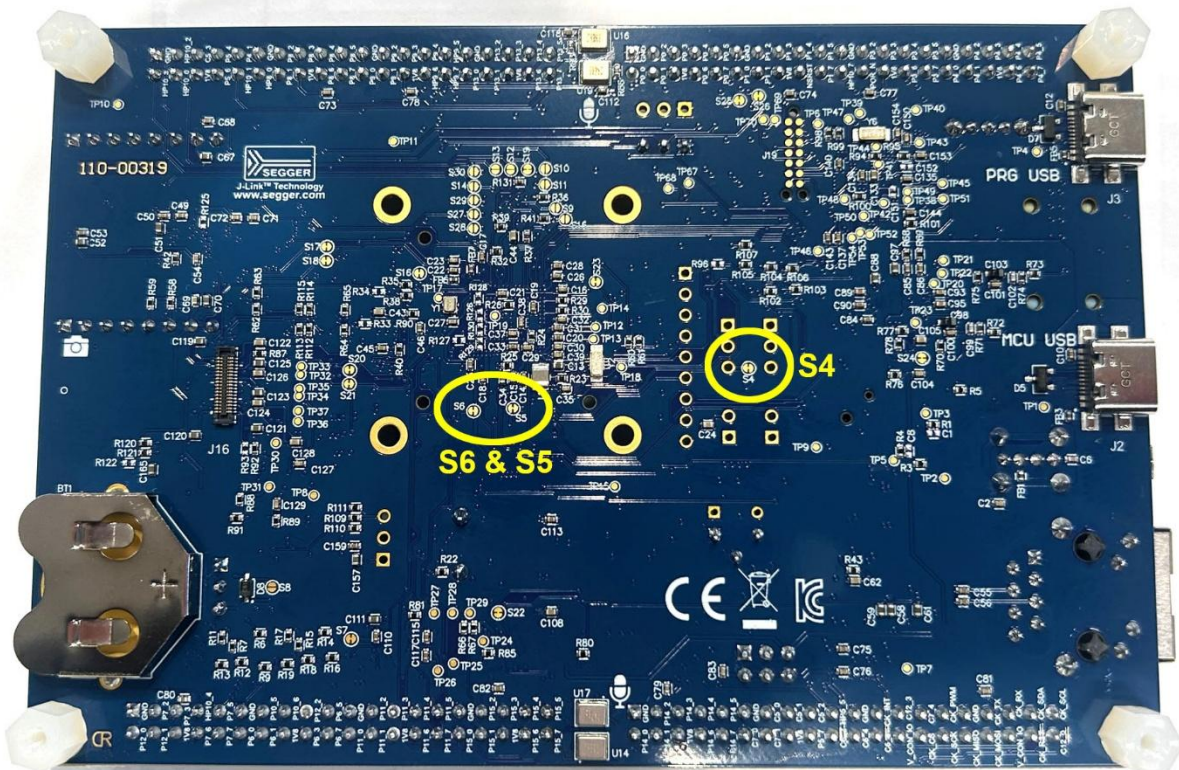
### Board Modifications

The DK-E8 Rev. B does not have JP1 through JP4 installed in the factory, the jumpers are bypassed with cuttable shunts as shown in the schematic. Before measuring power on the DevKit through these jumpers, it is necessary to cut the traces on S4 through S7.

At a minimum: Only cut S5 and measure current through JP2 (BUCK+MAIN+BATT).

Optionally: Cut S6 and connect a second ammeter to JP3 (VDD\_IO\_1V8).

When using USB: Cut S4 and measure USB PHY current through JP1 (VDD\_USB\_3V3).



## Document History

| Version | Change Log             |
|---------|------------------------|
| 1.0     | Initial public release |